

ECP FY17 AD Project Review

ECP Co-Design Project (WBS 1.2.5.3.04)

Center for Efficient Exascale Discretizations (CEED)



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EXASCALE COMPUTING PROJECT



U.S. DEPARTMENT OF
ENERGY

Office of
Science

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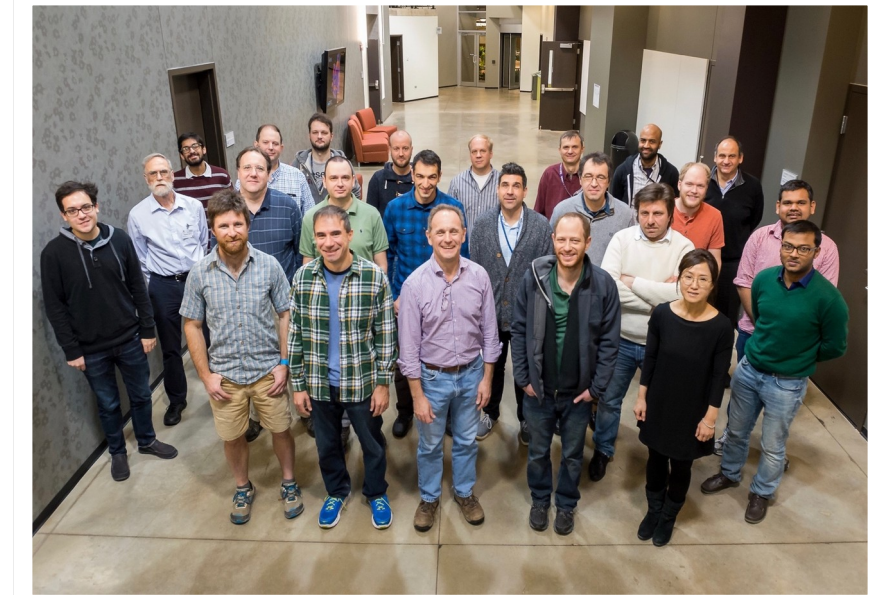
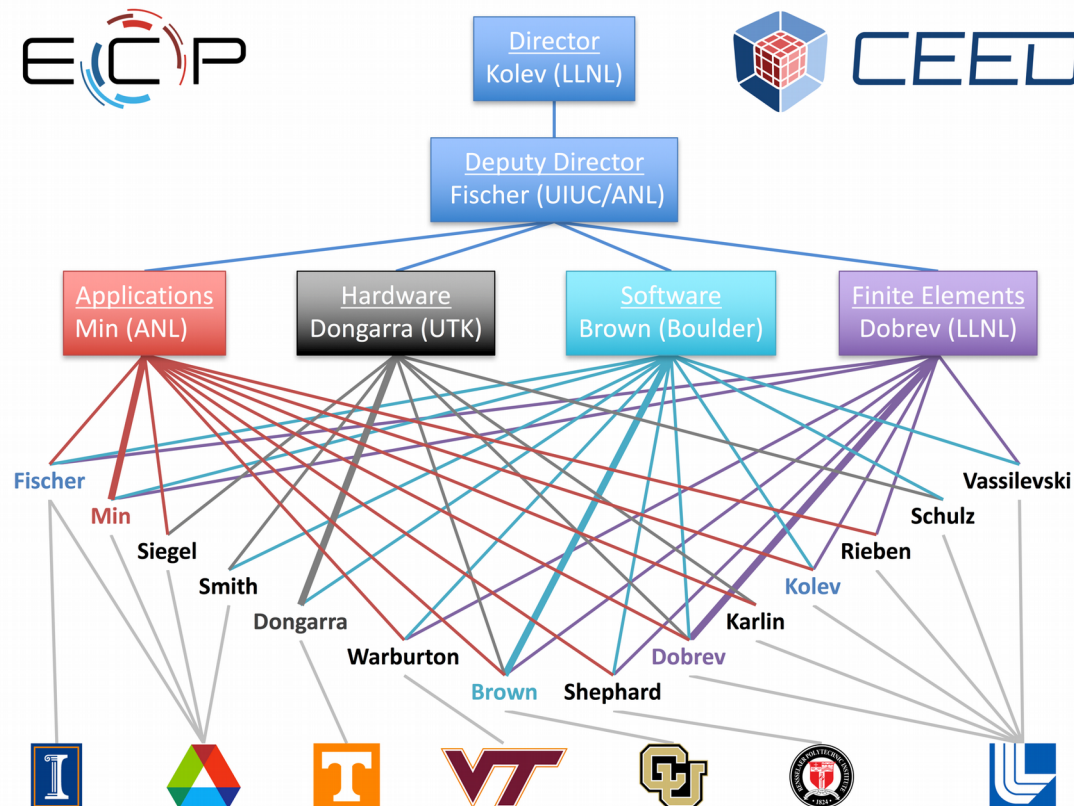
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EXASCALE
COMPUTING
PROJECT

Goals & Team

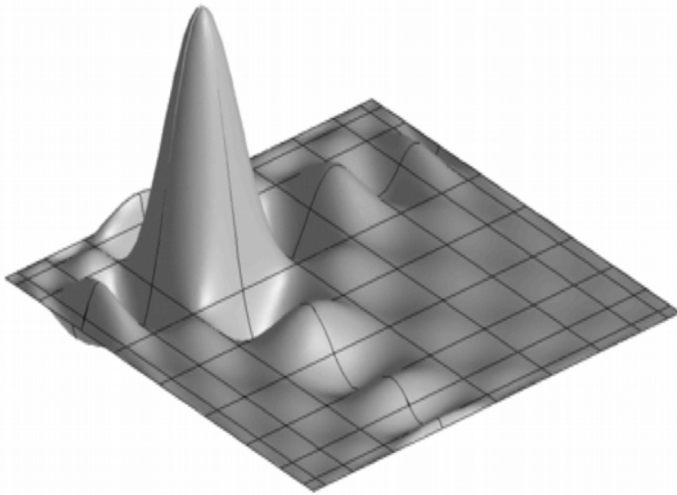
- CEED is focused on the *development of next-generation discretization software and algorithms to enable efficient simulations for a wide range of science applications on future HPC systems.*
- Funding: \$3.0M/year, 2 labs (LLNL, ANL), 5 universities



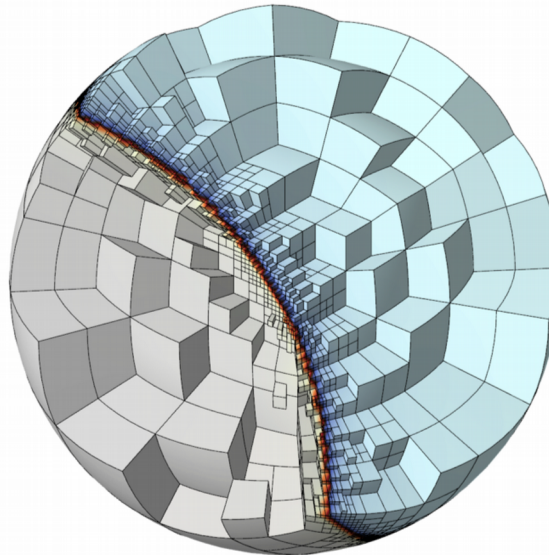
30+ researchers

Co-design Motifs

- PDE-based simulations on **unstructured grids**
- **high-order** and **spectral** finite elements
 - ✓ *any order space on any order mesh* ✓ *curved meshes,*
 - ✓ *unstructured AMR* ✓ *optimized low-order support*



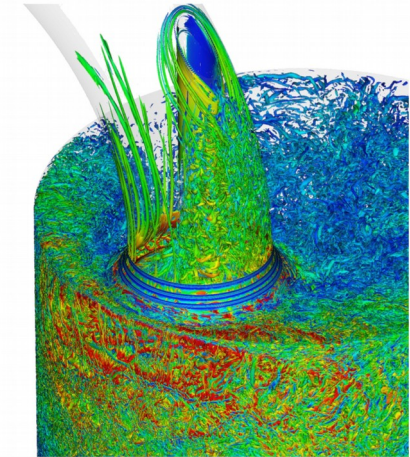
10th order basis function



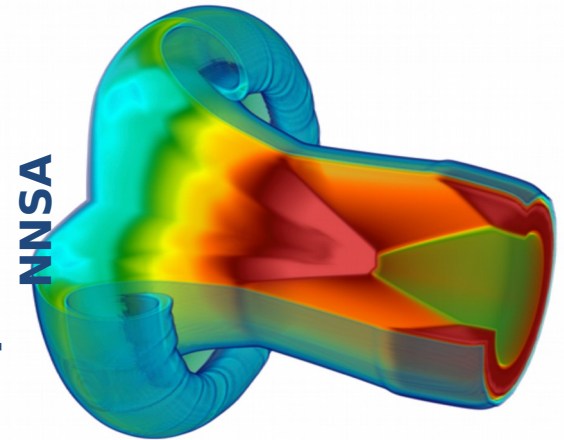
non-conforming AMR, 2nd order mesh

Project Overview

compressible FEM / NNSA / incompressible SEM / SC

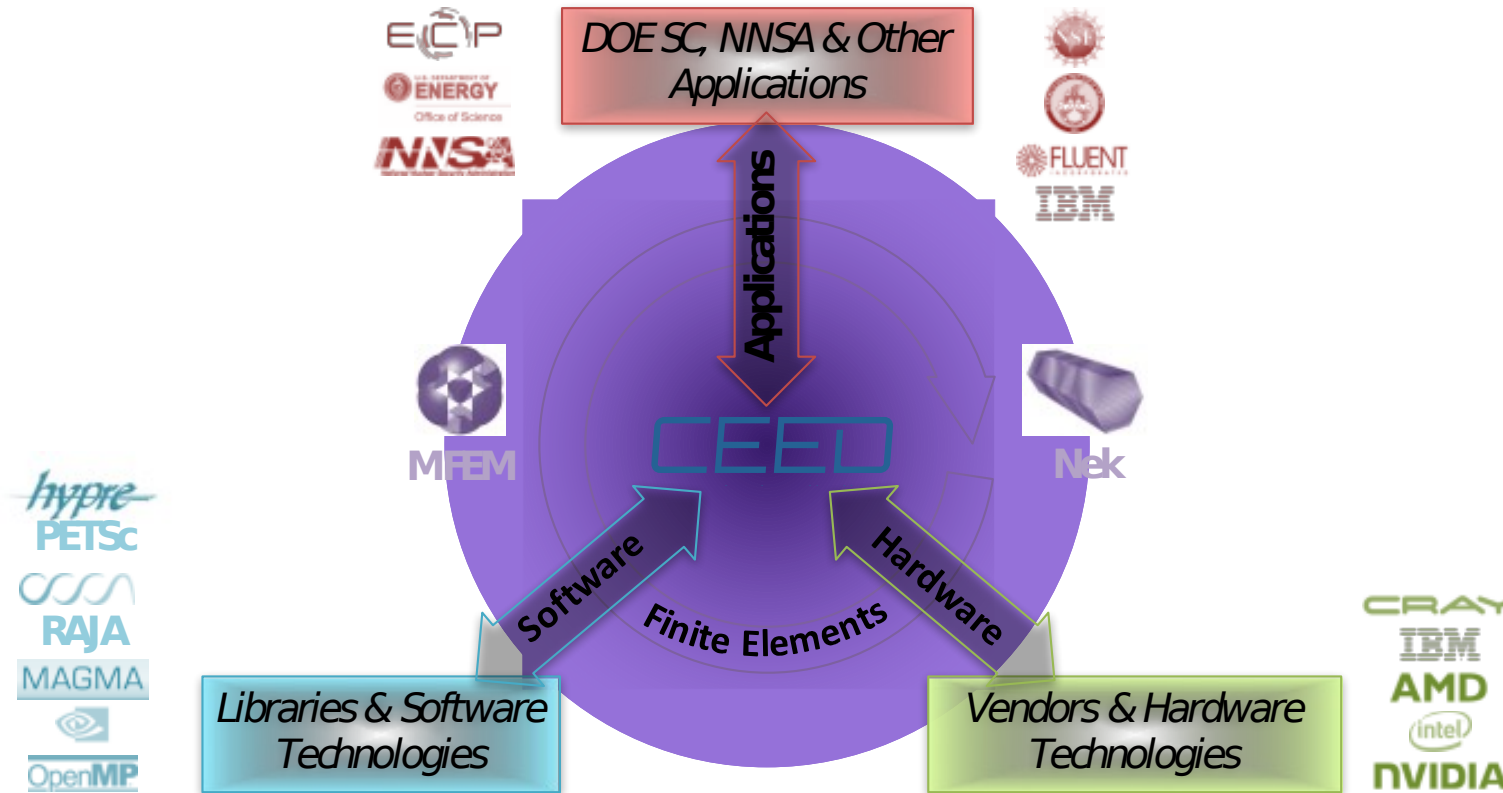


6th order DNS turbulence (Nek)



2nd order compressible shock hydro (MFEM)

R&D Thrusts



Four R&D thrusts focused on ECP customers

Co-Design Objectives:

- Help **applications** bridge the gap to exascale hardware
- Learn from & inform **vendors** about high-order methods

CEED users benefit from the combined efforts of:

- Nek5000
- MFEM
- MAGMA
- PETSc
- PUMI
- OCCA
- Holmes
- gslib

FY17 Milestones

- ❑ **Q2:** Select the first wave of ECP/CEED **applications** (applications targeting Nek or MFEM) and appoint **application liaisons**; team integration: project kickoff, online collaboration space; team integration: funding, staffing, milestones tracking
 - delivered **CEED-MS1 (EPIC)**, CEED-MS2 and CEED-MS3
- ❑ **Q3:** Engage **vendors** with initial **miniapps**; develop public-facing project website; meeting to discuss common finite element framework and kernels; formulate/document initial **bake-off (benchmark) problems**.
 - delivered CEED-MS4, CEED-MS5 and **CEED-MS6 (EPIC)**
- ❑ **Q4:** Perform **technology comparison** (bake-off) between the high-order and spectral element methods on application-motivated model problem; initial **integration** of CEED's discretization libraries in the selected first wave CEED applications; CEED first annual meeting @ LLNL.
 - working on CEED-MS7, **CEED-MS8 (EPIC)** and CEED-MS9

FY18 – FY20 Milestones

- ❑ **FY18:** **Q1.** Initial CEED API; **Q2.** Public release of CEED 1.0; Update CEED in first wave applications – showcase benefits; **Q3.** Propose high-order mesh/data format; **Q4.** Performance tuning of CEED software and first wave applications; CEED annual meeting.
 - milestones CEED-MS10 – CEED-MS21, 4 EPICs

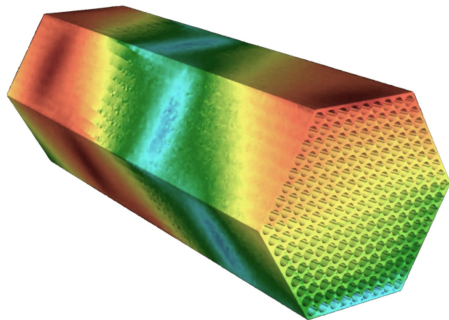
- ❑ **FY19:** **Q1.** Updated CEED API; engage second wave ECP/CEED applications; **Q2.** Public release of CEED 2.0; Update CEED in first wave applications – showcase benefits; **Q3.** Improved support for parallel AMR with load balancing in CEED software; **Q4.** Performance tuning of CEED software and 1st/2nd wave applications; CEED annual meeting.
 - milestones CEED-MS22 – CEED-MS33, 4 EPICs

- ❑ **FY20:** **Q1.** Engage third wave ECP/CEED applications; document performance improvements in CEED-enabled applications **Q2.** Public release of CEED 3.0; **Q3.** Efficient solvers for high-order discretizations; **Q4.** Document and popularize CEED-developed software and standards.
 - milestones CEED-MS34 – CEED-MS45, 4 EPICs

CEED Software Products

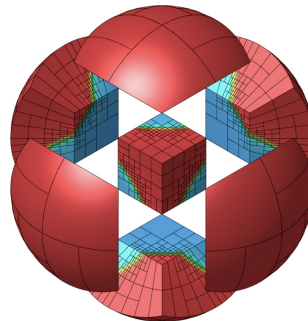
CEED's library model enables ECP apps to easily take advantage of the new discretization technologies

- state-of-the art CEED **discretization libraries**
 - ✓ better exploit the hardware to deliver significant performance gain over conventional methods
 - ✓ based on MFEM/Nek, low & high-level APIs



nek5000.mcs.anl.gov

High-performance spectral elements



mfem.org

Scalable high-order finite elements

CEED's proxies and general purpose libs target ECP vendors, STs, broader community

- **Ceedlings** - CEED kernels, bake-off probs & miniapps
 - ✓ main tools to engage vendors & external projects
- CEED **broadly applicable libraries**



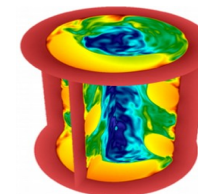
icl.cs.utk.edu/magma

LAPACK for GPUs, multi/many-core



libocca.org

Lightweight performance portability



GSLIB

PUMi

Parallel Unstructured Mesh Infrastructure

Holmes

PETSc

Main deliverable: all CEED software freely available on GitHub at <https://github.com/CEED>

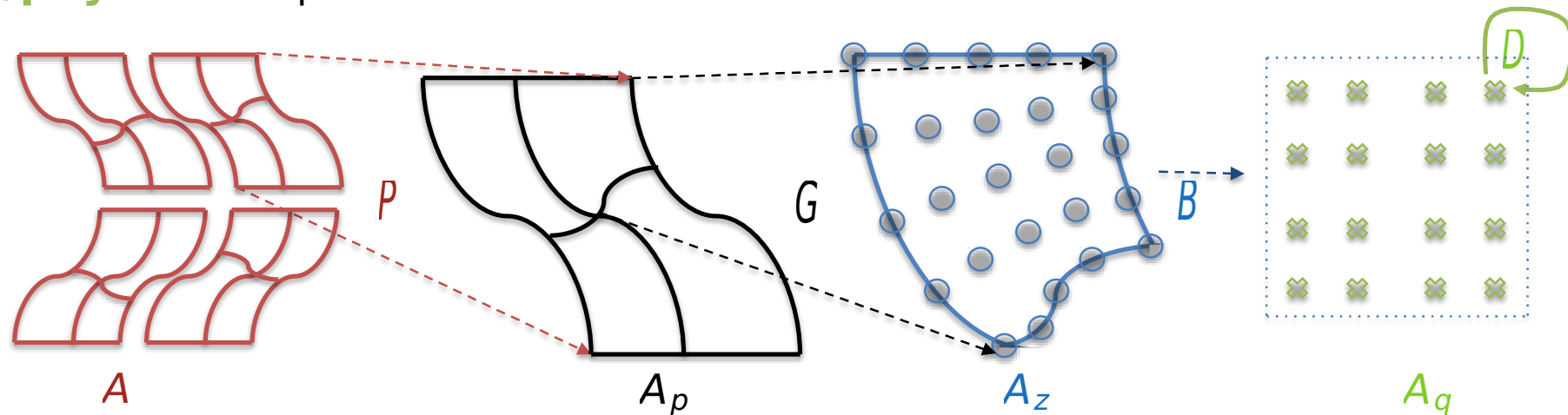
New releases: mfem-3.3, gslib, Laghos and NekCEM ceedling, ...

Efficient Operator Evaluation

Mathematical framework for our computational motif

$$A = P^T G^T B^T D B G P$$

Finite element operators (matrices) can be decomposed into **parallel**, **mesh topology**, **basis**, and **geometry/physics** components:



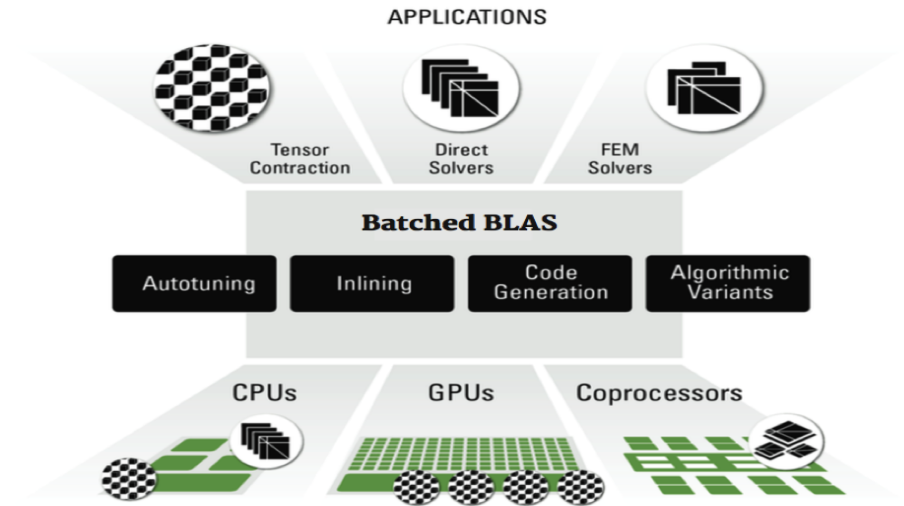
- **partial assembly** = store only D , evaluate B (tensor-product structure)
- better representation than A : *optimal memory, near-optimal FLOPs, hierarchical parallelism*
- *better mathematical algorithms trump code optimizations*

Batched Computing Technology

- Matrix-free basis evaluation needs efficient tensor contractions, e.g.,

$$C_{i1,i2,i3} = \sum_k A_{k,i1} B_{k,i2,i3}$$

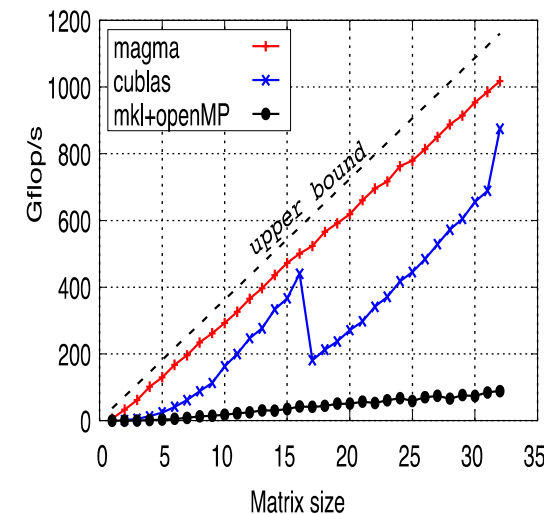
- CEED/MAGMA** designed batched methods to split the computation in many small high-intensity GEMMs, grouped together (batched) for efficient execution:



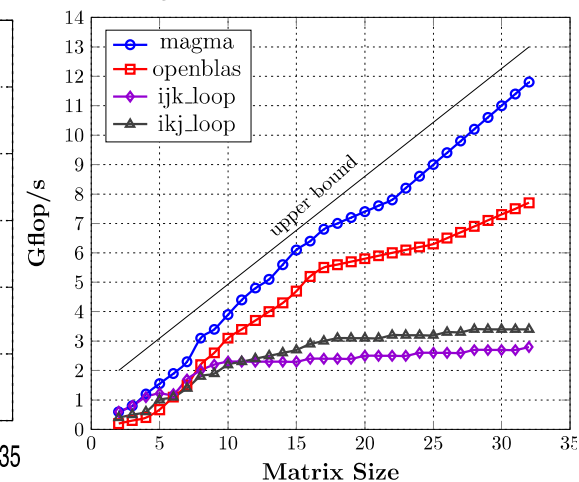
Batch_{ C_{i3} = A^T B_{i3}, for range of i3 }

- Developed techniques needed for autotuning, code inlining, code generation (reshapes, etc.), algorithmic variants for different architectures.
- Achieve 90+% of theoretically derived peaks.
- Significantly outperform vendor libraries.
- Released through MAGMA.

Batched DGEMMs on GPU
(P100, 100K)



Batched DGEMMs on ARM
(Tegra X1 : 4-core Cortex A57)



Batched Computing Technology

Targeted hardware platforms: need standardization to increase impact & handle implementation challenges



Memory hierarchies	Intel Haswell E5-2650 v3	Intel KNL 7250 DDR5 MCDRAM	IBM Power 8	ARM Cortex A57	Nvidia P100	Nvidia V100
	10 cores 368 Gflop/s 105 Watts	68 cores 2662 Gflop/s 215 Watts	10 cores 296 Gflop/s 190 Watts	4 cores 32 Gflop/s 7 Watts	56 SM 64 cores 4700 Gflop/s 250 Watts	80 SM 64 cores 7500 Gflop/s 300 Watts
REGISTERS	16/core AVX2	32/core AVX-512	32/core	32/core	256 KB/SM	256 KB/SM
L1 CACHE & GPU SHARED MEMORY	32 KB/core	32 KB/core	64 KB/core	32 KB/core	64 KB/SM	96 KB/SM
L2 CACHE	256 KB/core	1024 KB/2cores	512 KB/core	2 MB	4 MB	6 MB
L3 CACHE	25 MB	0...16 GB	8 MB/core	N/A	N/A	N/A
MAIN MEMORY	64 GB	384 16 GB	32 GB	4 GB	16 GB	16 GB
MAIN MEMORY BANDWIDTH	68 GB/s	115 421 GB/s	85 GB/s	26 GB/s	720 GB/s	900 GB/s
PCI EXPRESS GEN3 X16 NVLINK	16 GB/s	16 GB/s	16 GB/s	16 GB/s	16 GB/s	300 GB/s (NVL)
INTERCONNECT INFINIBAND EDR	12 GB/s	12 GB/s	12 GB/s	12 GB/s	12 GB/s	12 GB/s

Memory hierarchies for different type of architectures

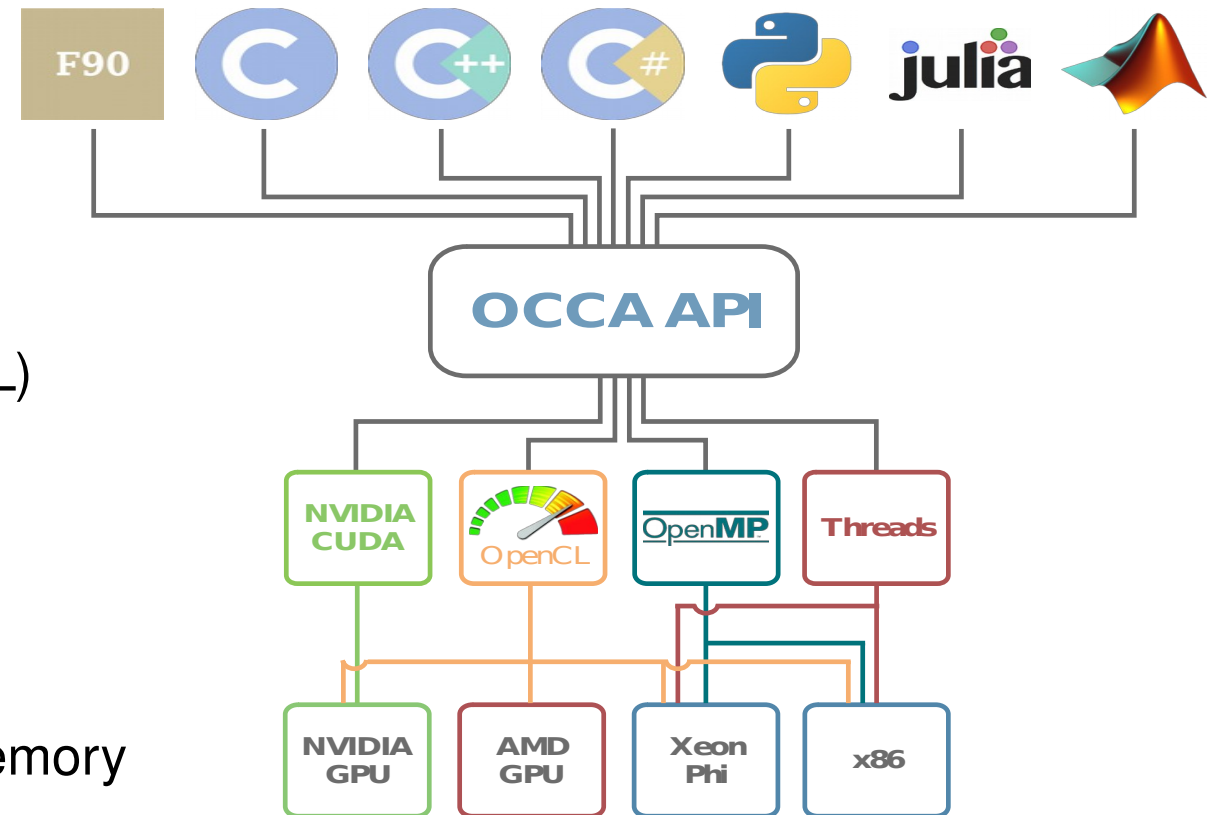
- MAGMA team lead crosscutting co-design efforts to define a Batched BLAS API standard with vendors and the broader community. Organized “Workshop on Batched, Reproducible, and Reduced Precision BLAS”.
- Batched tensor contractions relevant to convolutional neural networks for ML.

Lightweight Performance Portability

CEED/OCCA is an open-source library that provides an unified API for programming different types of devices, including CPUs, GPUs, Intel's Xeon Phi, FPGAs.

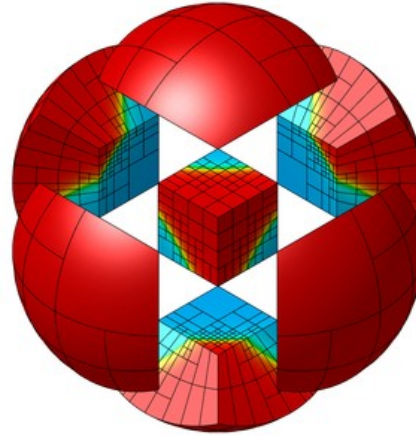
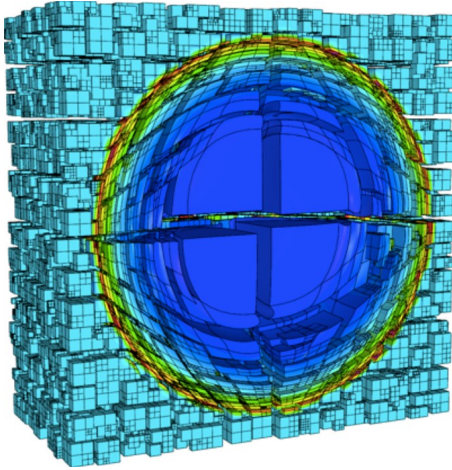
Features:

- Supported on many languages, such as C++, C, and Fortran
- *JIT compilation for kernels*
- Single kernel language for all backends (OKL)
- Currently supports Serial, OpenMP, CUDA, and OpenCL backends. Works with MPI
- MIT License, <http://www.libocca.org>
- Extensible backend API, allowing for future features. For example, support for unified memory in CUDA and mapped memory in OpenCL

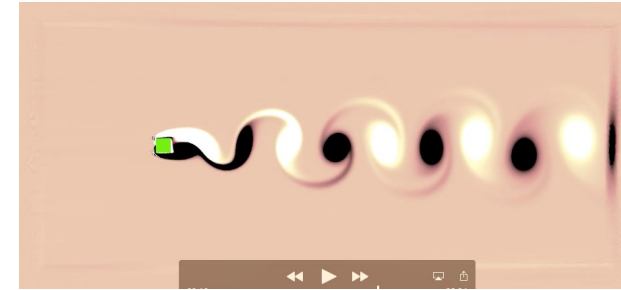


Lightweight Performance Portability

OCCA is being integrated and used *internally* in the following CEED software



MFEM



Holmes

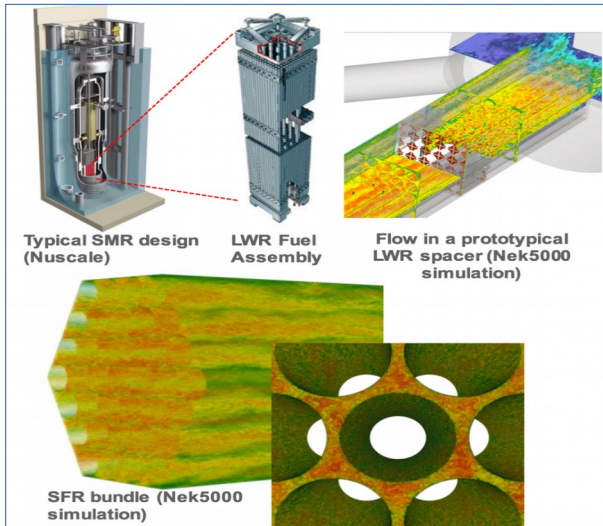
Key Benefits:

- Easy R&D of different evaluation kernels (e.g. different for different devices, orders, mixing C++ and Fortran)
- JIT technology key to enable compiler optimization of low-level HO loops (for variable order)

CEED-MS1 Summary

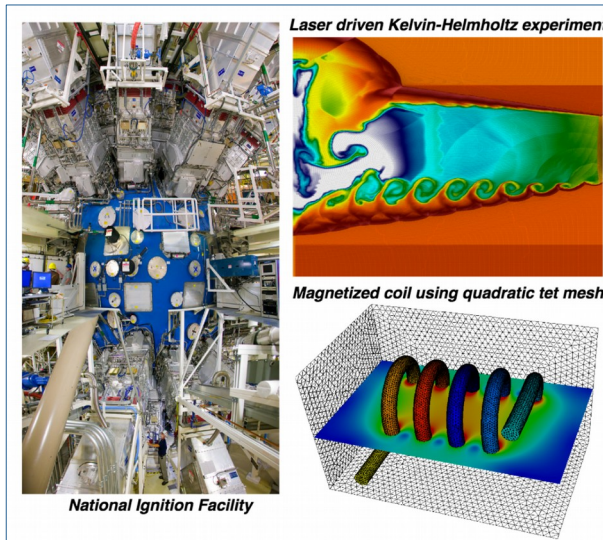
- Select 1-2 first wave ECP/CEED applications.
- Appoint liaisons on the CEED team for the selected first wave applications.
- Co-design a discretization interface with the selected applications.
- Outline an integration plan for CEED technologies into first wave applications
- Report summarizing the chosen applications and the planned CEED engagements.

Engaged First Wave ECP/CEED Applications



ExaSMR: Coupled Monte Carlo Neutronics and Fluid Flow Simulation of Small Modular Reactors.

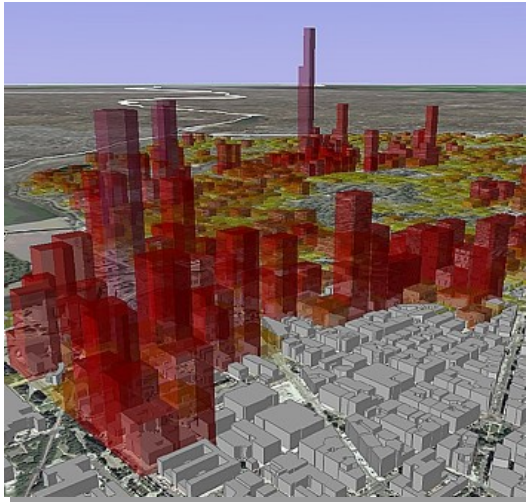
- ❑ Goal: Coupled thermal-hydraulics/neutronics analysis
- ❑ Require scalable simulations up to one billion spectral elements and trillions of grid points; complex mesh structures, external and internal fluid mixing, advanced RANS/LES models, efficient time stepping, and improved parallel mesh I/O support.
- ❑ *Application liaison: Elia Merzari (ANL)*
- ❑ **Shared milestone: Nek5000 porting on GPUs, KNL performance**



MARBL: Next-gen Multi-physics Simulation Code.

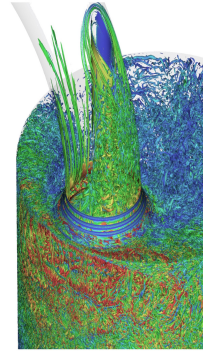
- ❑ Goal: Modular physics simulation capabilities with increased performance portability and flexibility.
- ❑ Require single-fluid multi-material hydrodynamics and radiation/magnetic diffusion simulation; inertial confined fusion, pulsed power experiments, and equation of state/material strength analysis
- ❑ *Application liaison: Vladimir Tomov (LLNL)*
- ❑ **Shared milestone: Developing Lagrangian hydro miniapp**

Other ECP Applications

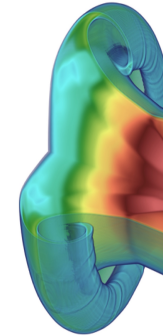


Urban systems

learn & inform **vendors** about high-or

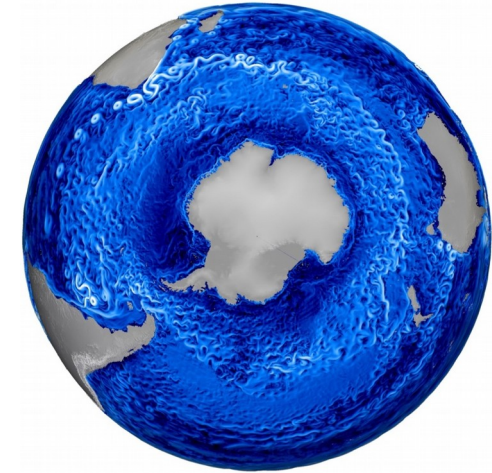


6th order DNS turbulence (Nek)



2nd order compressible

**Combustion
(Nek5000)**



**Climate
(ACME)**

In CEED-MS1 we also explored opportunities for future engagements with applications in the second and third waves of the ECP/CEED applications:

- ❑ **Urban:** Multi-scale Coupled Urban System (ANL)
- ❑ **Combustion:** Transforming Combustion Science and Technology with Exascale Simulations (SNL)
- ❑ **ACME:** Cloud-Resolving Climate Modeling of the Earth's Water Cycle (SNL)

CEED-MS6 Summary

- Identify initial **miniapps** (CEEDlings) that represent sub-components of interest from the selected 1st wave of apps.
- Candidate miniapps include Nekbone, MFEM's HPC examples and a new Lagrangian miniapp for MARBL
- Identify a set of **bake-off (or benchmark) problems (BPs)** that either represent part of a miniapp or are of general interest. BPs solve a simple physical problem, including discretization, linear or nonlinear solve, and potentially a small number of time steps.
- Identify **kernels** - algorithms, usually written as a single function that does not call other functions - representing one step in a BP or a miniapp, that is important for their overall performance.
- Use miniapps, BPs and kernels to **engage vendors and ST projects**: both to educate them about the requirements of high-order methods (push) and to learn how to improve their performance (pull).
- Deliver through the CEED website: ceed.exascaleproject.org and GitHub repositories: github.com/CEED.

CEED Miniapps

- **Nekbone**: A lightweight subset of Nek5000; solves a standard Poisson equations; weak-scaled to 6 million MPI ranks. Currently support OpenACC/CUDA-based GPU variants.
- **[New!] Laghos**: A proxy for a sub-component of the MARBL; solves the time-dependent Euler equation of compressible gas; makes use of unstructured moving meshes.
- **[New!] NekCEM CEEDling**: A NekCEM miniapp; solves the time-dependent Maxwell equations; supports OpenACC-based GPU variant.
- **Holmes**: An experimental testbed for multi-level parallel implementations of high-order finite element computations; under development.
- **HPGMG-FE**: A full multigrid solver for a third-order accurate finite element discretization.

Miniapps freely available on GitHub at: <https://github.com/CEED>

More info at: <http://ceed.exascaleproject.org/miniapps>

CEED Bake-Off Problems

- CEED's *bake-off problems* are high-order kernels/benchmarks designed to test and compare the performance of high-order codes.
- Objective – Assess performance across:
 - ❑ Key kernels for PDE-based simulation.
 - ❑ A variety of architectures, network configurations, node counts, etc.
 - ❑ Suite of fast, production-level, implementations (both internal and external to CEED).
- Net result will be for ***all*** CEED-enabled applications to benefit from the fastest implementations across each of the exascale (and other HPC) platforms.
- First set of BPs (BP1–BP4) were developed at CEED kick-off meeting and have been documented in the CEED-MS6 report as well as at <http://ceed.exascaleproject.org/bps>.
- BPs already picked-up by external high-order community: https://github.com/kronbichler/ceed_benchmarks_dealii

CEED Bake-Off Problems

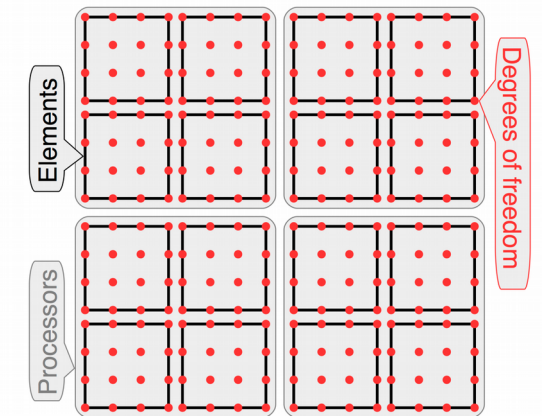
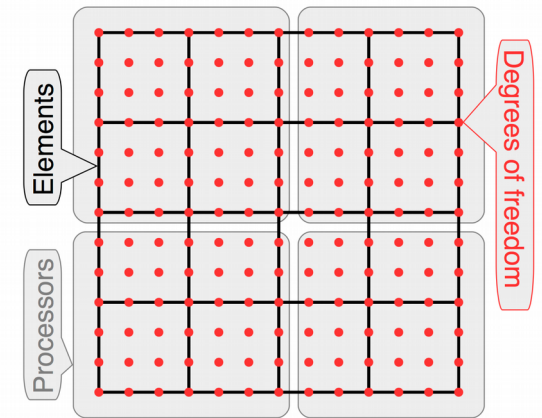
BP1: Solve $\{Bu=f\}$, where $\{B\}$ is the mass matrix.

BP2: Solve the vector system $\{Bu_i=f_i\}$ with $\{B\}$ from BP1.

BP3: Solve $\{Au=f\}$, where $\{A\}$ is the Poisson operator.

BP4: Solve the vector system $\{Au_i=f_i\}$ with $\{A\}$ from BP3.

- Range of polynomial orders: $\{p=1, 2, \dots, 8\}$, at least.
- Cover range of sizes: from 1 element/MPI rank up to the memory limit.
- BP1 and BP2 are relevant for many hyperbolic substeps in transport problems. BP3 and BP4 reflect pressure, momentum, and diffusion updates in fluid/thermal transport.
- Vector forms BP2 and BP4 reveal benefits of increased *data reuse* and of *amortized communication overhead*.
- Benchmark repo: <https://github.com/CEED/benchmarks>



**BP terminology: T-
and E-vectors of HO
dofs**



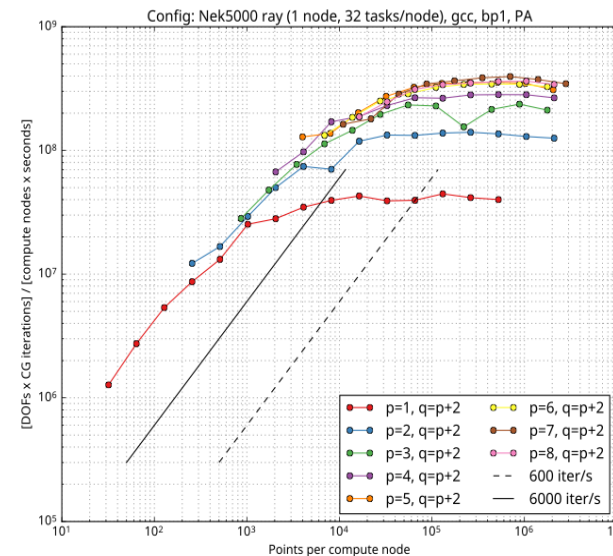
CEED Bake-Off Problems

- Initial results on BG/Q, KNL, GPU testbed, show similar performance for MFEM and Nek5000.
- Performance roll-off with reduction in local problem size indicates strong-scale limit, i.e. min problem size per node/ max number of nodes for a problem of fixed size
- Operators on vector fields improve scalability.
- Compiler can affect performance significantly, e.g. xIC on P100 testbed, Intel on KNL.

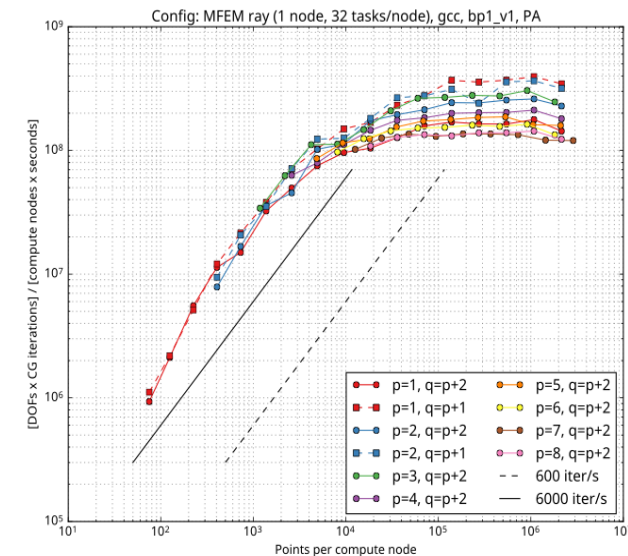
Max performance, strong limit values on diff. machines

Machine/Problem	r_{max} (MDOFs)	$n_{\frac{1}{2}}$ (KDOFs)	$t_{\frac{1}{2}}$ (ms)
Ray/BP1-gcc	400/350	15/35	0.075/0.200
KNL32/BP1-intel	200/150	15/30	0.150/0.400
KNL32/BP1-gcc	100/100	20/8	0.400/0.160
Vulcan/BP1-gcc	35/40	15/10	0.857/0.500
KNL32/BP2-intel	200/100	30/15	0.300/0.300
KNL32/BP2-gcc	100/100	50/15	1.000/0.300
Vulcan/BP2-gcc	35/30	21/10	1.200/0.666
Vulcan/BP3-gcc	/30	/10	/0.666
Vulcan/BP3-xlc	/20	/10	/1.000
Vulcan/BP4-gcc	/20	/10	/1.000

Performance vs. local problem size for various polynomial orders, p



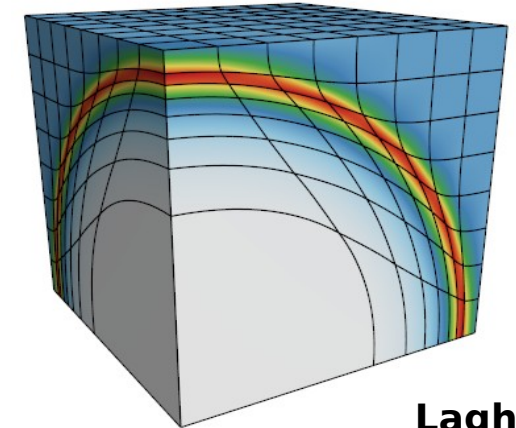
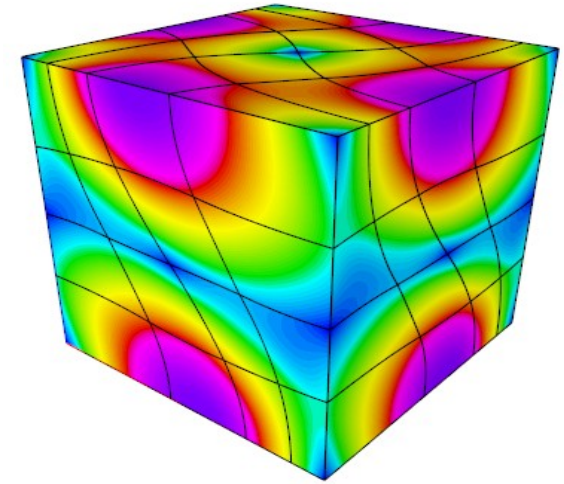
(a) Nek5000



(b) MFEM

Laghos: LAGrangian High-Order Solver

- Laghos is a new MFEM-based miniapp for high order hydrodynamics: Captures the basic structure of many other compressible shock hydro codes, including MARBL.
- Continuous and discontinuous high-order finite element discretization spaces of runtime-specified order.
- Moving (high-order) curved meshes.
- Supports full assembly and partial assembly options.
 - Clear separation between assembly and point-based computations.
- Supports 2D and 3D unstructured meshes, with quadrilateral and hexahedral elements.
- C++ code with domain-decomposed MPI parallelism.
 - Public release: <https://github.com/CEED/Laghos>



**Laghos
simulations:
Taylor Green
(top),
Sedov blast
(bottom)**

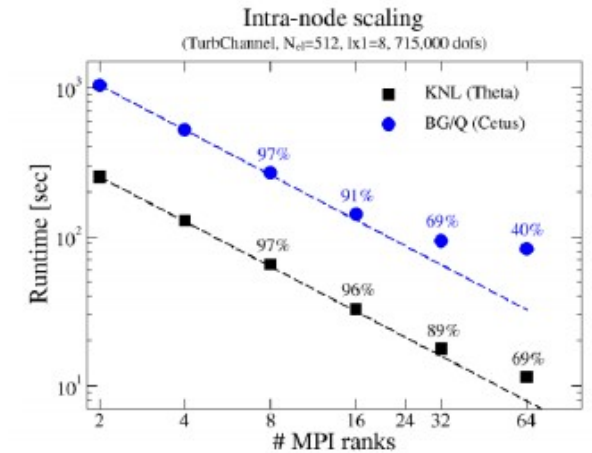
Integration of CEED in MARBL



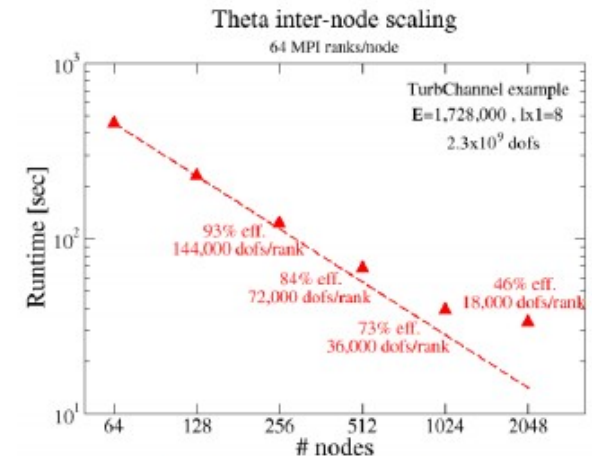
- Integration in MARBL/BLAST will be facilitated by initial integration in Laghos.
 - Both BLAST and Laghos are based on MFEM.
 - Improvements in Laghos can readily be extended to BLAST.
- GPU integration (based on OCCA) in Laghos is close to completion.
- Reorganization of some MARBL computations is already guided by Laghos.
- Laghos had immediate impact on MARBL's development process.
 - Easier introduction of new people (e.g. vendors, developers, summer students).
 - Development of new methods without complicated code navigation.

Integration of CEED in ExaSMR

- Evaluating Nek performance on three problems of interest
 - Single rod case.
 - 2x2 bundle with mixing vanes.
 - 17x17 bundle without mixing vanes (representative of SMR designs).
- Progress in porting Nek5000 to GPUs
 - Participated in OpenACC hackathon (with CEED).
 - Ported hot kernels of the code with a ~40x speed-up (2.5s GPU vs 97s CPU).
 - Current peak at ~240 Gflops in double precision, aiming at ~1Tflops.
 - Currently extending to full code.
- Continued work on KNLs.
 - Preliminary results for scaling on Theta.



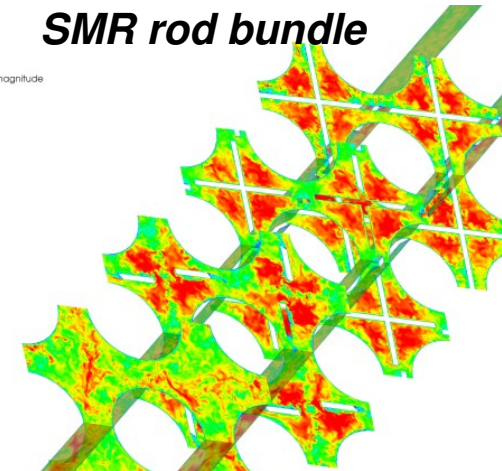
(a) Intranode strong scaling, (blue) BG/Q Mira vs (black) KNL Theta.



(b) Strong scaling up to 2048 nodes.

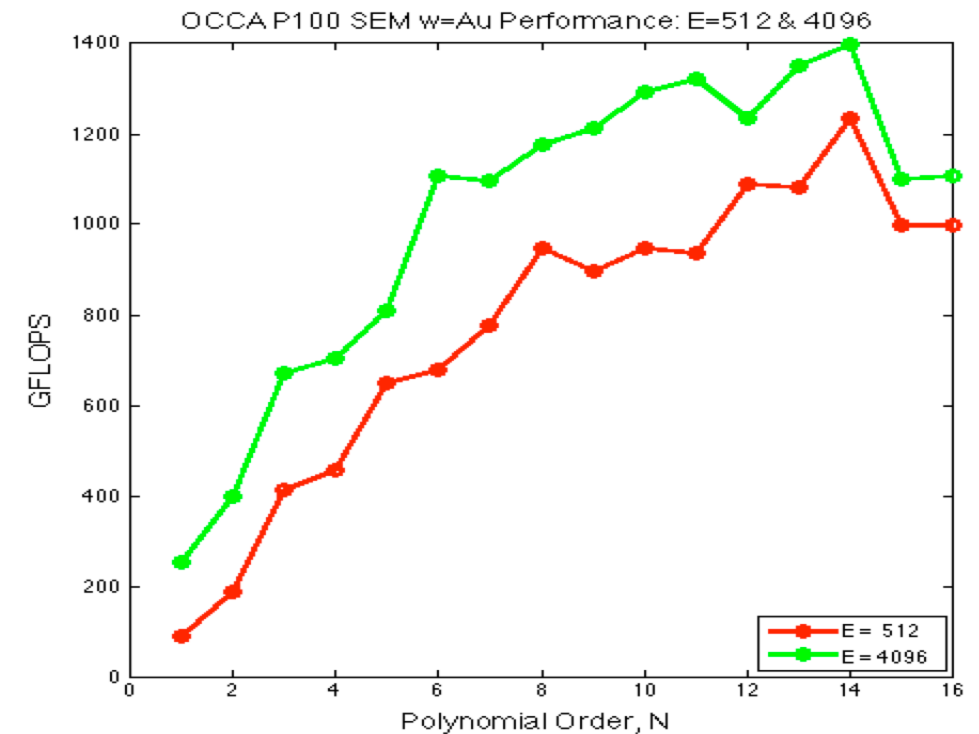
Status of GPU-port for Nek5000

- Shared milestone with SMR
- Initial OpenACC port for $P_N P_N$ formulation (fully tested) includes
 - ❑ CG and GMRES solvers for velocity and pressure
 - ❑ Hybrid Schwarz-multigrid preconditioner
 - ❑ Dealiased advection operator
 - ❑ Initial kernels are realizing 100-250 Gflops on P100 (Pascal) nodes (64-bit precision).
- A parallel effort with **OCCA-based kernels** (Virginia Tech) is realizing > 1.2 Tflops.
 - ❑ Plan is to use OpenACC as initial port, followed by adaptation of OCCA-generated kernels.
 - ❑ ***CEED objective is to attain the highest performance possible.***
- Porting strategies for GPUs will apply across most accelerator-based architecture.



OCCA Performance on P100

- VTech developed highly-tuned OCCA code for central kernel delivering $>1.2\text{TFLOPs}$, with weak dependence on polynomial order $N=8-14$ (**good**).
- Performance drops 30% when number of elements decreases from 4096 to 512.
 - 70% efficiency when going from 1 million points per node ($E=4096$, $N=8$) to $\frac{1}{4}$ million points/node (**bad**).
 - We are analyzing the ramifications for overall turn-around time.



Engagements with Vendors, STs and Other Projects

- Interacting with **AMD** through **Laghos**
- Attended **Intel** hackathon, run key kernels on their **simulator**
- **ARM, Cray, Intel, AMD, IBM** attending CEED 1st annual meeting
- Working with **MPICH** team on low-latency lightweight-MPI
- **STRUMPACK** and **PETSc** integrated in **MFEM**
- CEED software stack builds together with **Spack**
- Future work with **OpenMPI, ALPINE/VTK-m, SNLApp, KokkosKernels, Zfp, SOLLVE**
- **MFEM** working with **IntelligentLight** on high-order visualization
- **PUMI** and **MFEM** part of several funded **SciDAC** fusion proposals
- **MFEM** joining **xSDK**, part of new **FASTMath** proposal

Collaboration with MPICH

CEED is working with the MPICH team to explore improved scaling strategies, including low-latency *lightweight-MPI*.

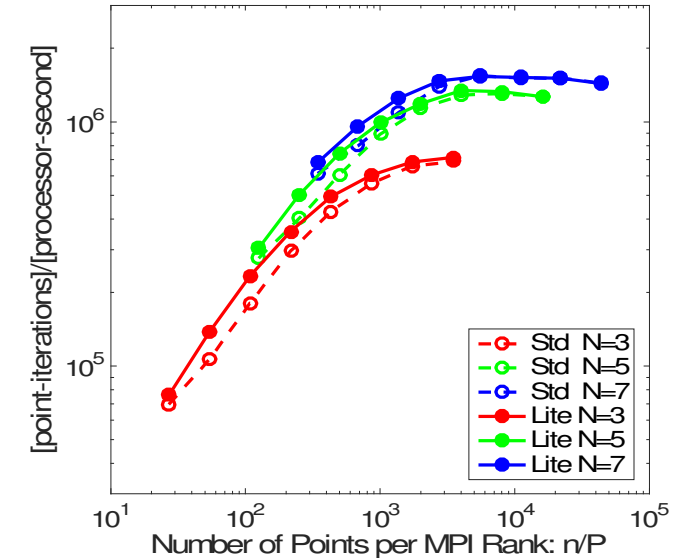
Findings:

- ❑ Reductions in communication overhead, specifically latency, at the strong-scale limit allows utilization of more processors.
- ❑ Under certain strong-scale (fixed power) conditions, lightweight-MPI can deliver a 1.6 reduction in time-to-solution.

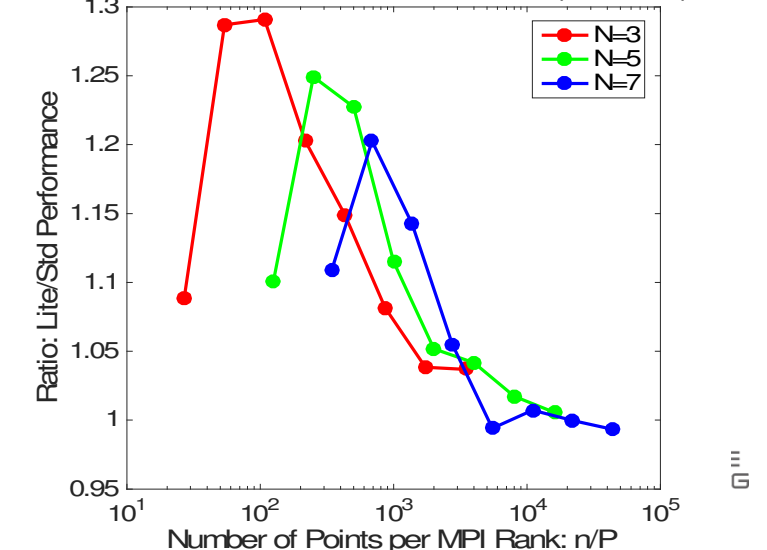
Joint publication with the MPICH team accepted at SC17:

Raffenetti et al.,
Why Is MPI So Slow? Analyzing the Fundamental Limits in Implementing MPI-3.1,
 SC'17, November 2017, Denver, Colorado USA

Nek5000 Mass-Matrix Inversion, Lite & Std MPI.



Nek5000 Mass-Matrix Inversion (Lite/Std)



Project Costs

Institution	Annual Budget	% Spent	FY18 carryover
LLNL	\$1275K	65%	\$209K (16%)
ANL	\$775K	47%	\$153K (20%)
VTech	\$200K	47%	\$45K (23%)
UTK	\$200K	57%	\$22K (11%)
CU Boulder	\$200K	10%	\$150K (75%)
UIUC	\$175K	33%	\$72K (41%)
RPI	\$175K	58%	\$44K (25%)
Total	\$3000K	\$1580K (53%)	\$795K (27%)

- Project started Q2 FY17
- Jan 2017 – Jul 2017 = 58%
- LLNL post-doc and new staff hires are starting in August
- CU Boulder student starting in Fall

Upcoming Milestones

FY18 Q1

- **Initial CEED API (libCEED)**; Prepare CEED distribution; Performance bottlenecks in 1st wave apps.

FY18 Q2

- **Public release of CEED 1.0**; Update CEED in first wave applications – showcase benefits; Update miniapps.

FY18 Q3

- **Propose high-order mesh/data format**; Improved unstructured AMR; Engage low-order apps.

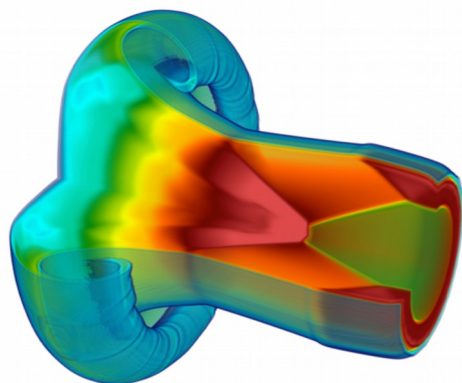
FY18 Q4

- **Performance tuning of CEED software and first wave applications**; Tet meshes; CEED annual meeting.

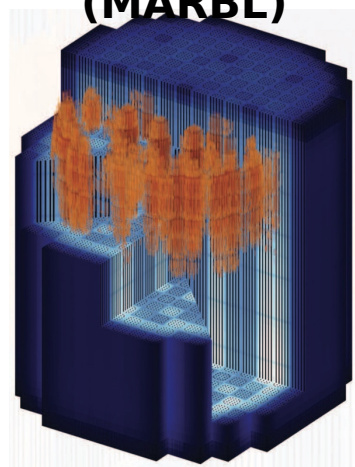
FY19 Q1

- **Engage 2nd wave apps**; Document performance improvements; Update CEED API.

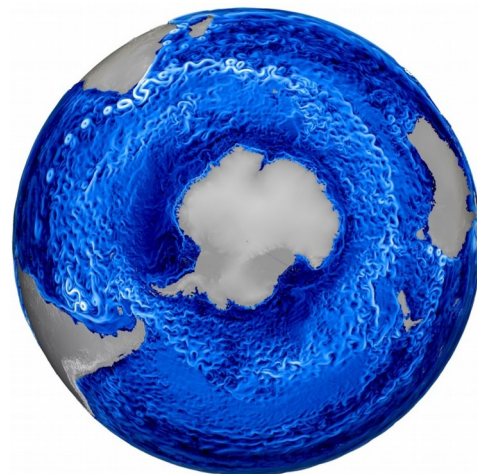
1st, 2nd and 3rd Wave of Applications



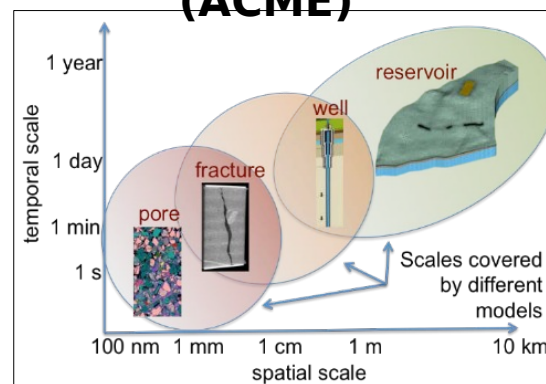
**Compressible flow
(MARBL)**



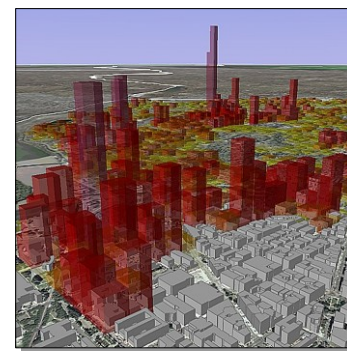
**Modular
Nuclear
Reactors
(SMRs)**



**Climate
(ACME)**



**Subsurface
(GEOS)**



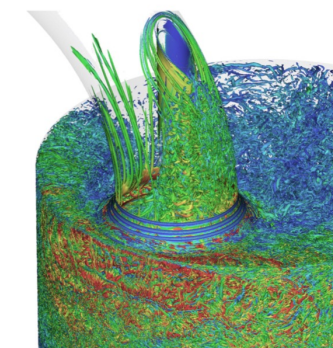
**Urban systems
(Urban)**



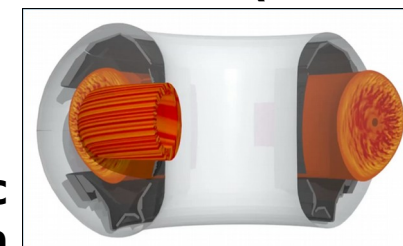
**Additive
Manufacturing
(ExaAM)**



Wind Energy

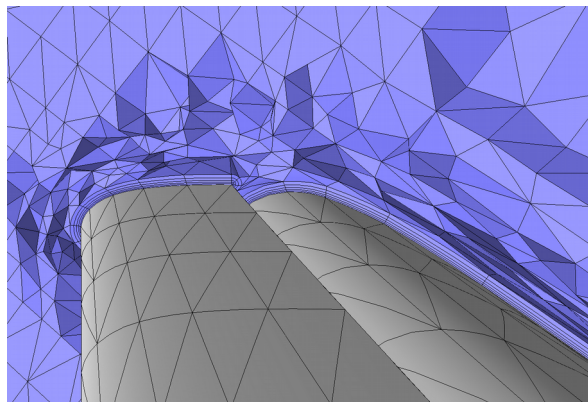


Combustion (Nek5000)

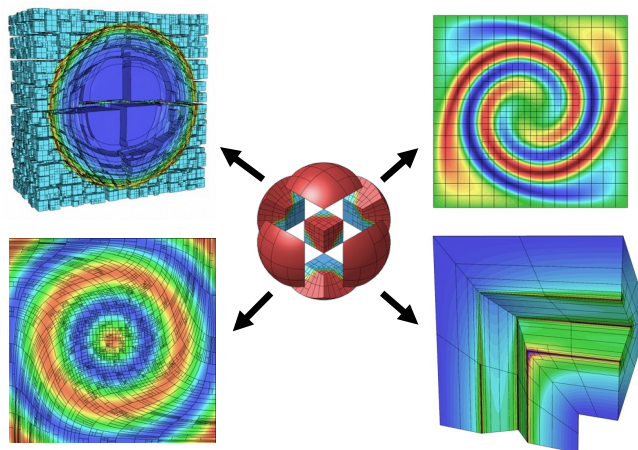


**Magnetic
Fusion
(WDMApp)**

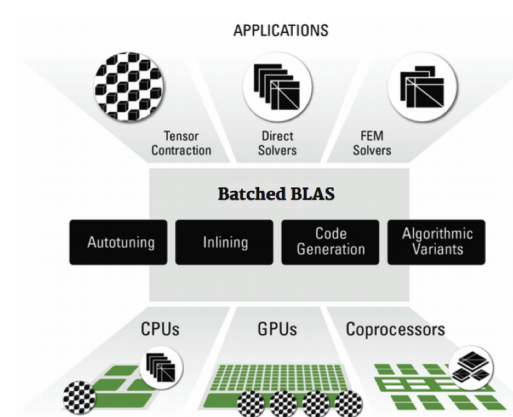
Future R&D / HO software ecosystem



High-order Meshes



Unstructured AMR



Batched dense tensor contractions

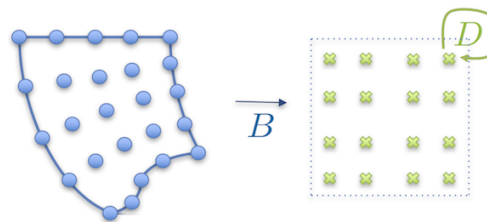
PETSc



Scalable matrix-free solvers

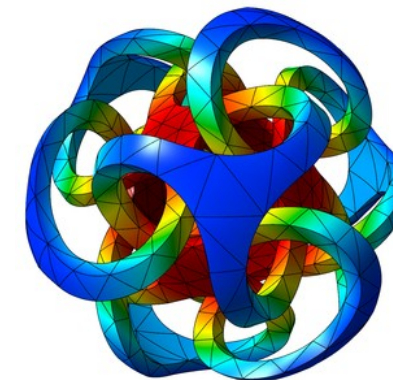
(Partners: PETSc, hypre, KAUST, ...)

$$A = P^T G^T B^T D B G P$$



Efficient High-Order Operator Format

(Partners: SNLApp, KokkosKernels, ...)
More info at: <http://ceed.exascaleproject.org/fe>



High-Order Visualization
(Partners: ALPINE/VTK-m, ACME, ExaWind, IntelligentLight, ...)

Project Risks and Challenges

- Maintaining team focus within the 3.0M/year funding level.
- NDA concerns about vendor announcement / engagement for CEED's university team members.
- Reviewing new software for open-source release at NNSA labs delays public release. One of the challenges is lack of ECP-wide policy on open-source software licensing.
- Guidelines on engaging external (non-ECP) community: KAUST, deal.ii, iLight, SciDAC?